COMSCI / ECON 206 | FINAL RESEARCH PROJECT | AUTUMN 2026

Trust the Card, Not the Resemblance

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A0 landscape 1189 × 841 mm. Submit PPTX + PDF. Commit poster PDF/preview to GitHub and embed in ROOT README.md.

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KEY MESSAGE

On a misaligned card, a public forfeit of 5 makes (Trust, Keep) the unique equilibrium. A similarity label changes no score. Seven pairs did not follow that Keep prediction. Those rounds are mixed with order, so this is not a causal result.

01 Question and foundations

Research question & verified gap

What public rule makes Trust the equilibrium when breaking pays more?
Closest: Bracht & Feltovich 2008; Fehr & Rockenbach 2003; Giang et al. 2012.
The forfeit is printed on the card. The label cannot waive it.

Strategic model and Intellectual Merits

Supplier promises. Buyer decides. Both lock before either is shown.
Forfeit of 5 makes (Trust, Keep). Payoffs are (6, 6).
No forfeit: (Verify, Break). Payoffs are (2, 10).
Nashpy at b1380f7. The pairs do not follow the card.

Three complementary lenses

Game theory: (Trust, Keep) with it; (Verify, Break) without it.
Social choice: (2, 10) to (6, 6). Decider +4. Promiser −4.
Mechanism: the label cannot waive the printed forfeit.

Auction or allocation comparison

Bids choose the Promiser only. Forfeit equals the second bid.
Fixed 5 keeps (Trust, Keep). A second bid must exceed 4.
(8, 3) gives (Verify, Break). At 4, no unique equilibrium.
Not played. The comparison stops there.

02 Model and verified evidence

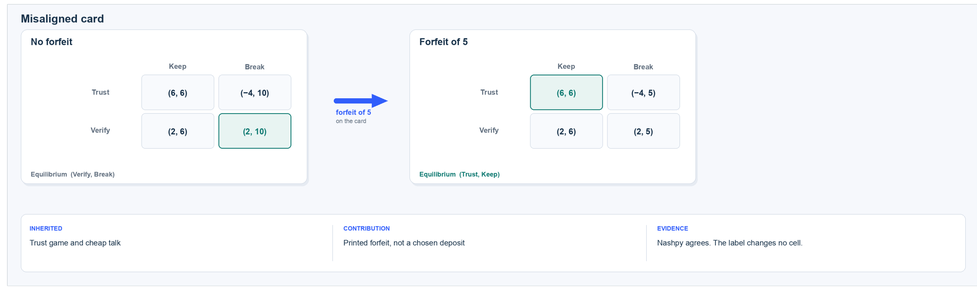


Figure 1. No forfeit: (2, 10). Forfeit of 5: (6, 6). Nashpy agrees.

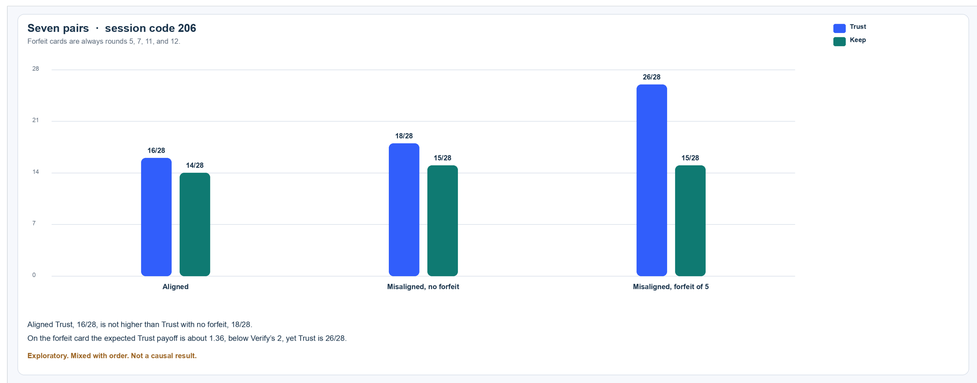


Figure 2. Forfeit cards are rounds 5, 7, 11, and 12.
Expected Trust payoff there is about 1.36, below 2, but Trust was 26/28.
Aligned card: Break pays 3, Keep pays 6, yet Keep is 14/28.
Commit b1380f7. Exploratory. No p-value.

03 Practical impacts and roadmap

Practical impacts and SDG targets

Collected forfeit of 5 flips the card to (Trust, Keep).
Buyer decides. Supplier pays if the forfeit is collected.
16.6: card, forfeit, and appeal visible before lock.
Baseline: no forfeit. Matrix only. Keep did not rise.
10.2: who can pay 5. Object in writing before collection.

Future research roadmap

Limit: Keep stays near half when Keep pays more.
Economics: why break? Noise, or doubt it is collected.
Zijie Wang and Lin Zhang may comment, if they agree.
Computing: random order, payoff check, collection line.
Behavior: at least 10 new pairs. None have been run.
If higher Keep is not more common, test a chosen escrow.

Integration and updated named contributions

Muhan Chen: the Promiser chooses; the forfeit is 5; the label is off the matrix.
Checks: dominance, Nashpy, and seed 206 (48, 24, 22, 18).
As the sole author, I am responsible for every section and deliverable. Ideas from peers and classroom contributors are in the acknowledgments.

OCT 12 Contribution lab and duet defense
OCT 14 Final symposium and innovation
OCT 21 Final submission at 23:00 Shanghai

Full acknowledgments

I thank Prof. Luyao Zhang for the PS1 final review of 25 September 2026, which required a two-sided game and visible matrices.
I thank Prof. Kenneth S. (Ken) Rogerson for asking us, as Program Discussant of the 7 September 2026 workshop, to say what practical decision the result could inform, and to keep a classroom result separate from a real-world effect.
I thank Han Zhang for asking me to make the comparison in the paper easier to see, and to say who would be affected by the rule.
Zijie Wang and Lin Zhang gave ideas in class.

Selected references

Bracht & Feltovich 2008, 10.1007/s00182-007-0092-3. Fehr & Rockenbach 2003, 10.1038/nature01474. Giang, Bell & Buchner 2012, 10.1371/journal.pone.0047809.

OPEN MATERIALS

huggingface.co/spaces/dku-comsci-econ206-2026/Trust_Forfeit
github.com/dku-comsci-econ206-Autumn2026/PS2-FP12-Muhan-Chen
Tested model commit b1380f7